

COMPUTER NETWORKS

Overview

This course introduces:

- ✓ Fundamental networking concepts, terminologies, and technologies used in data communication and computer networking.
- ✓ Course materials will be taught through class lecturers, individual textbook readings, discussions, assignments, and Practical labs
- ✓ Continuous Assessment Test will contribute to 50% (written CAT, Practical CAT, Assignments and presentations)
- Final written Exam will contribute to 50% of your results

Course Content

- Data Communications
- Data Link Layer
- Network Layer
- Transport Layer
- Application Layer

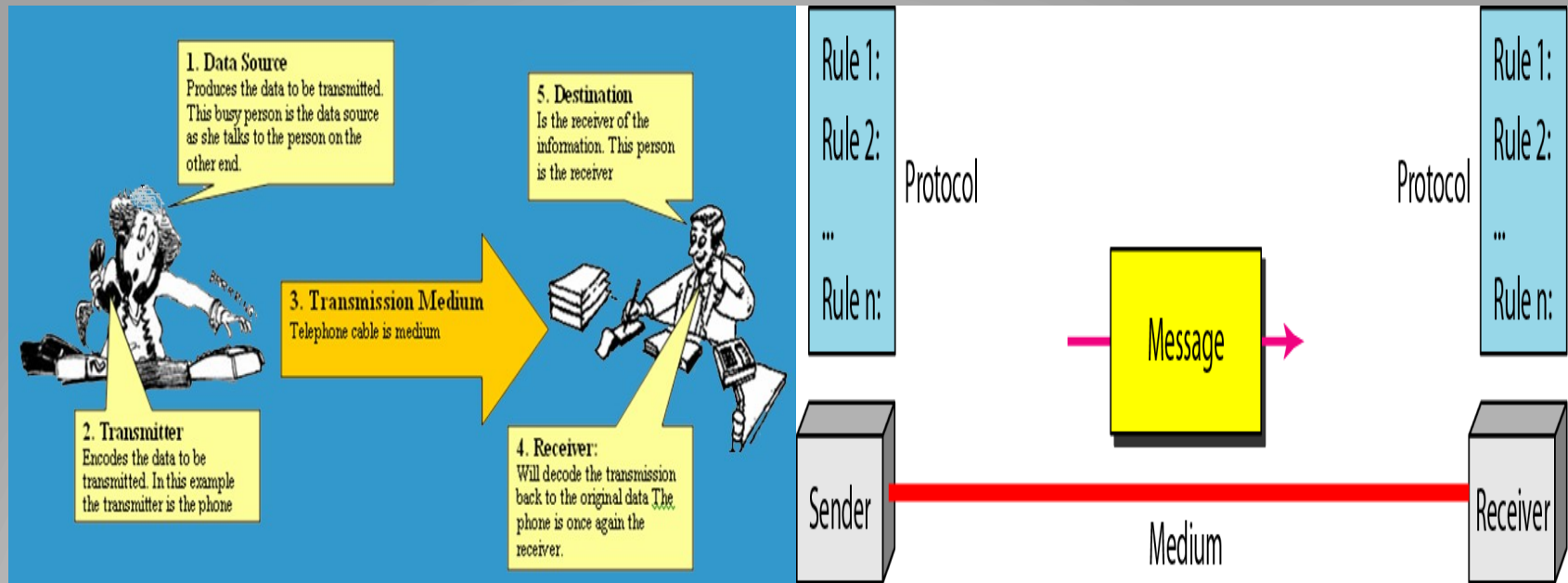
LAB:

- Configuration of Static Routing
- Configuration of Dynamic Routing: RIP
- Configuration of Dynamic Routing: EIGRP
- Configuration of Dynamic Routing: OSPF
- Switching: Clearing interconnected switches, VLANS, VTP....

DATA COMMUNICATION

- **Telecommunication** refers communication at a distance.
- The term **data** refers to **information presented in whatever form is agreed upon by the parties creating and using the data.**
- **Data comes** in different forms; **text, numbers, images, audio, and Video**
- **Data communications** are **the exchange of data between two devices** via some form of transmission medium such as a wire cable.

Components of Data Communication



1. Source/Destination (Protocol)
2. Transmitter (Sender)
3. Transmission Medium (Medium)
4. Receiver (Receiver)
5. Data (Message)

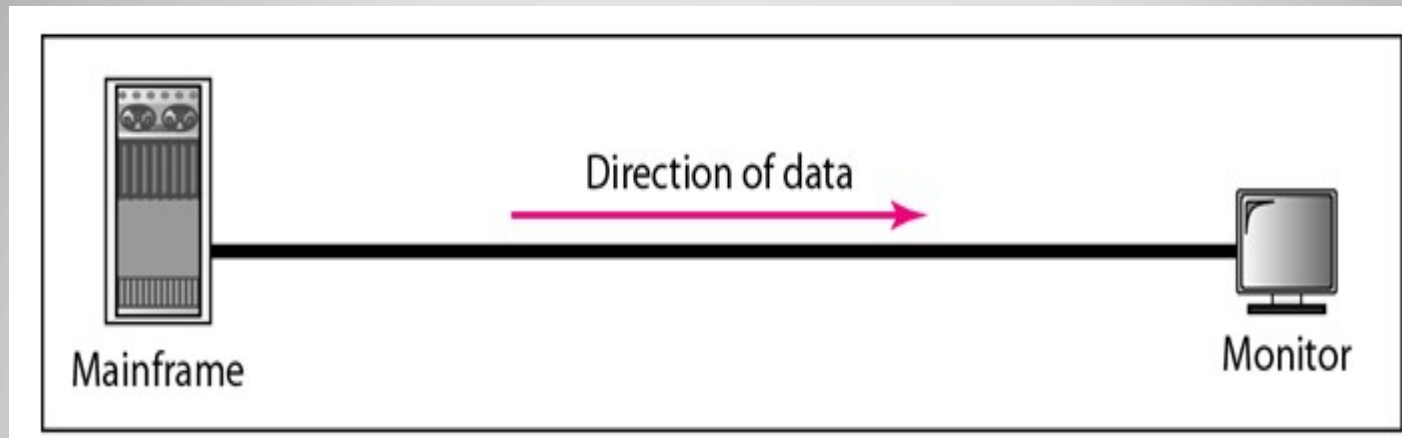
Direction of Data Flow

There are three directions of data flow;

1. Simplex
2. Half Duplex
3. Full Duplex

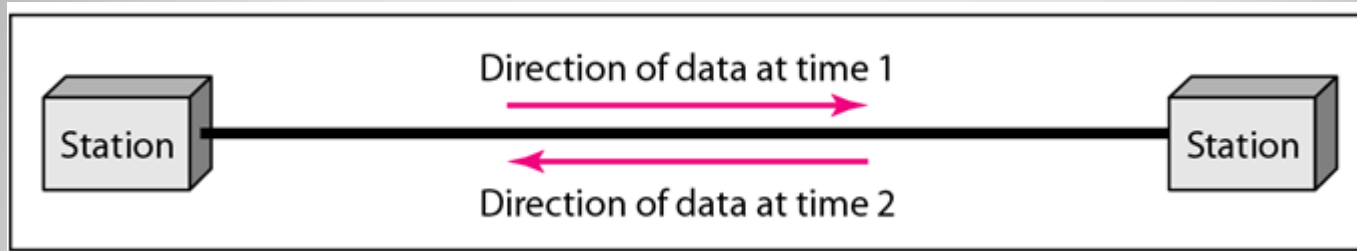
Simplex

- In simplex mode, **the communication is unidirectional, as on a one-way street.** Only **one of the two devices on a link can transmit**; the other can only receive
- Examples: Keyboards, traditional monitors



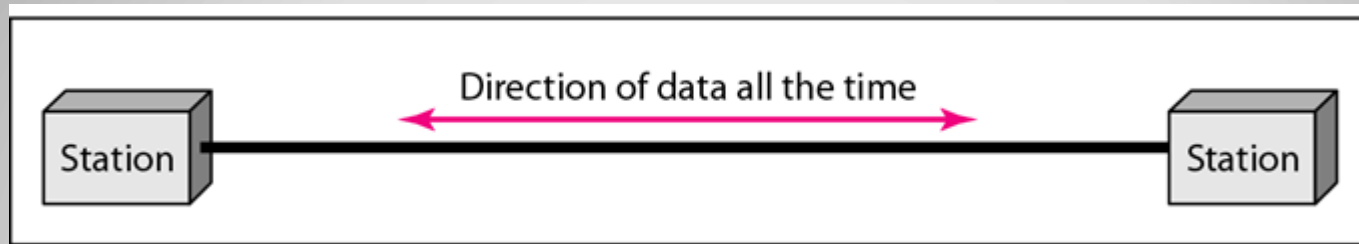
Half Duplex

- In half-duplex mode, each station can both transmit and receive, **but not at the same time.**
- When one device is sending, the other can only receive, and vice versa.
- Example: a walkie-talkie



Full Duplex

- In full-duplex mode(also called duplex), **both stations can transmit and receive simultaneously.**
- Example: landline



Networks

- A **network** can be defined **as a set of devices (Nodes) connected by communication links.**
- A node **can be a computer, printer, or any other device** capable of **sending and/or receiving data** generated by other nodes on the network.
- A link can be a cable, **air, optical fiber, or any medium** which can transport a signal carrying information.

Components and Categories of a Network

- A network is composed of different things like;
 1. end nodes (device + interface)
 2. Links
 3. interconnection network
 4. etc

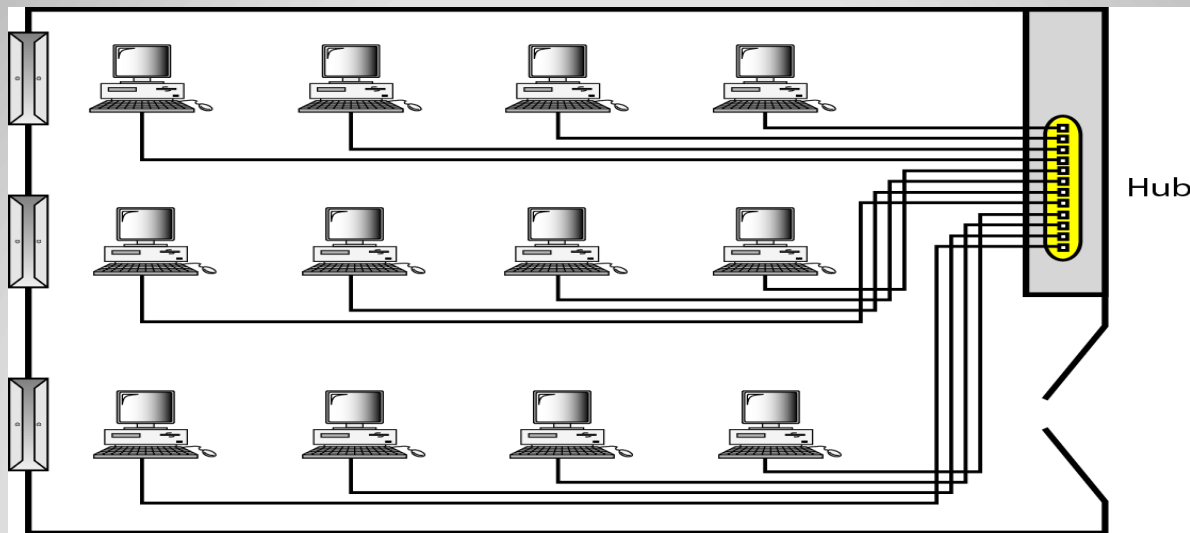
Network Categories

Network Categories include;

1. Local Area Networks (LANs)
2. Wide Area Networks (WANs)
3. Metropolitan Area Networks (MANs)

LAN

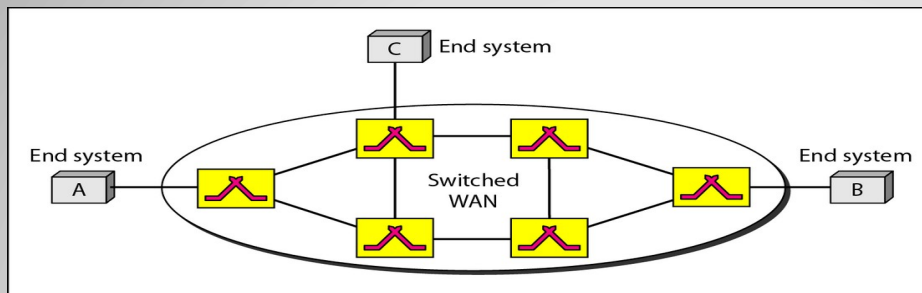
- **Local Area Networks (LANs):** LAN is usually privately owned and links the devices in a single office, building, or campus.
 - Short distances
 - Designed to provide local interconnectivity



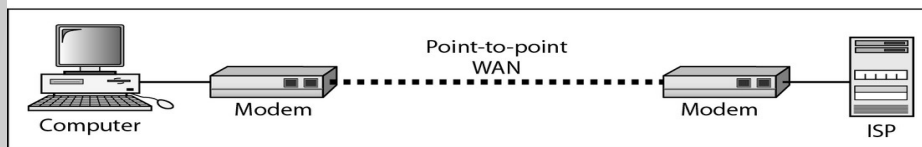
WAN

Wide Area Networks (WANs): WAN provides long-distance transmission of data, image, audio, and video information over **large geographic areas that may comprise a country, a continent**, or even the whole world.

- Long distances
- Provide connectivity over large areas



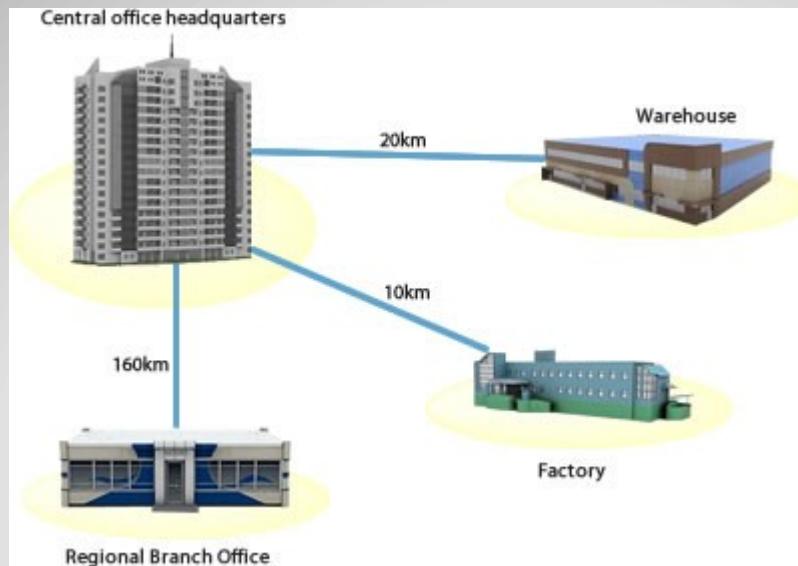
a. Switched WAN



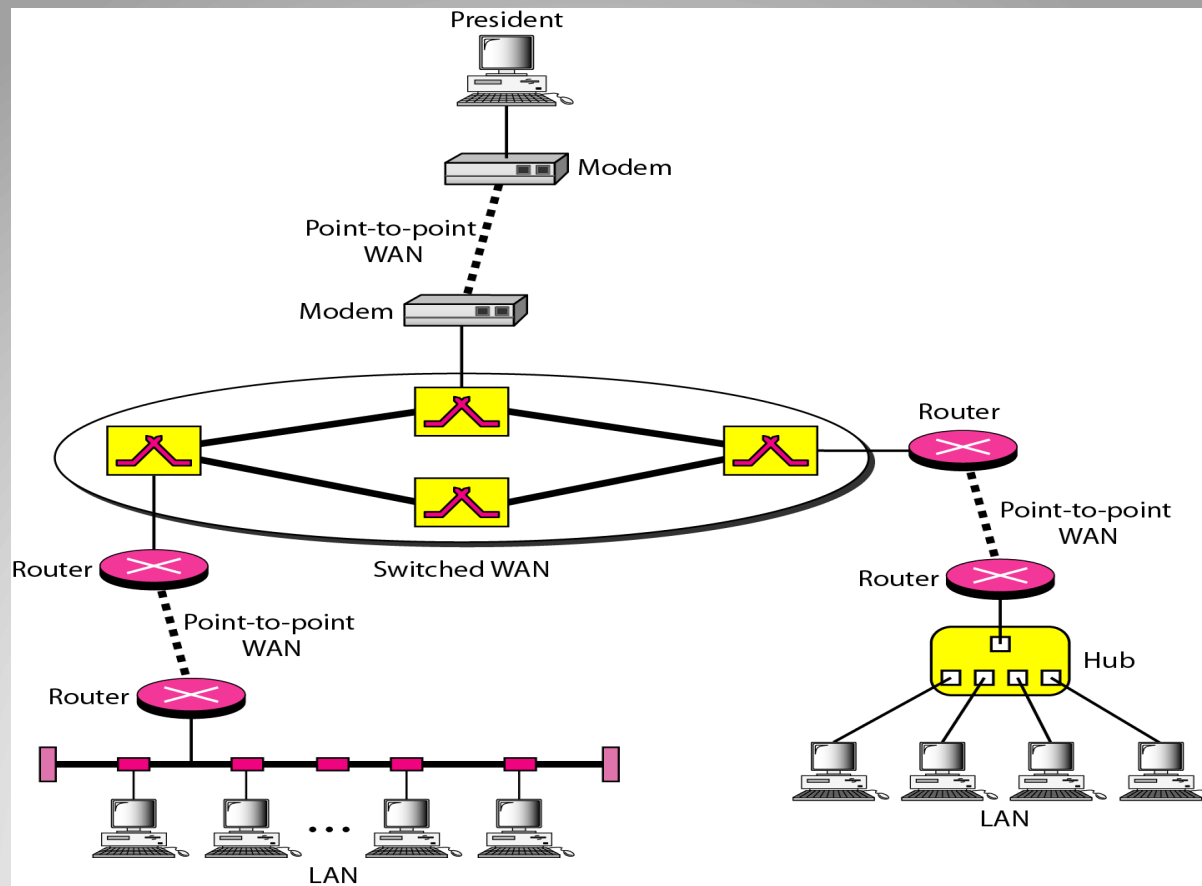
b. Point-to-point WAN

MAN

- **Metropolitan Area Networks (MANs):** MAN is a network with **a size between a LAN and a WAN.**
 - Provide connectivity over areas such as a city, a campus



A heterogeneous network made of four WANs and two LANs



Network Criteria

- A network must be able **to meet a certain number of criteria. The most important of these are:**

1. Performance:

- Depends on Network Elements
- Measured in terms of Delay and Throughput

2. Reliability:

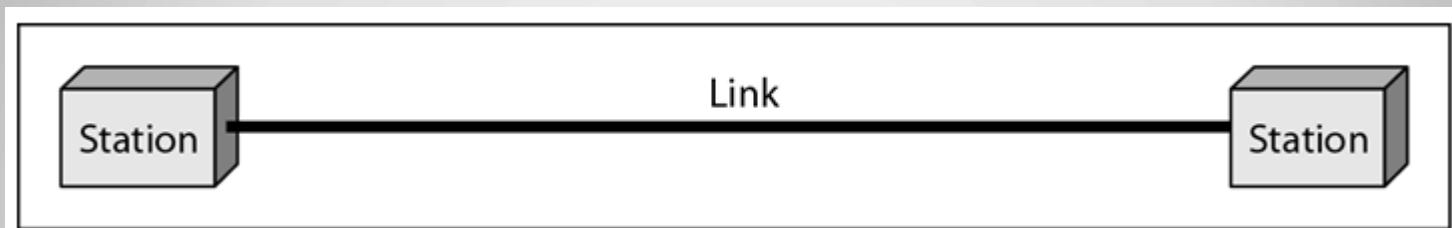
- Failure rate of network components
- Measured in terms of availability/robustness

3. security:

- Data protection against corruption/loss of data due to:
 - Errors
 - Malicious users

Types of Connection

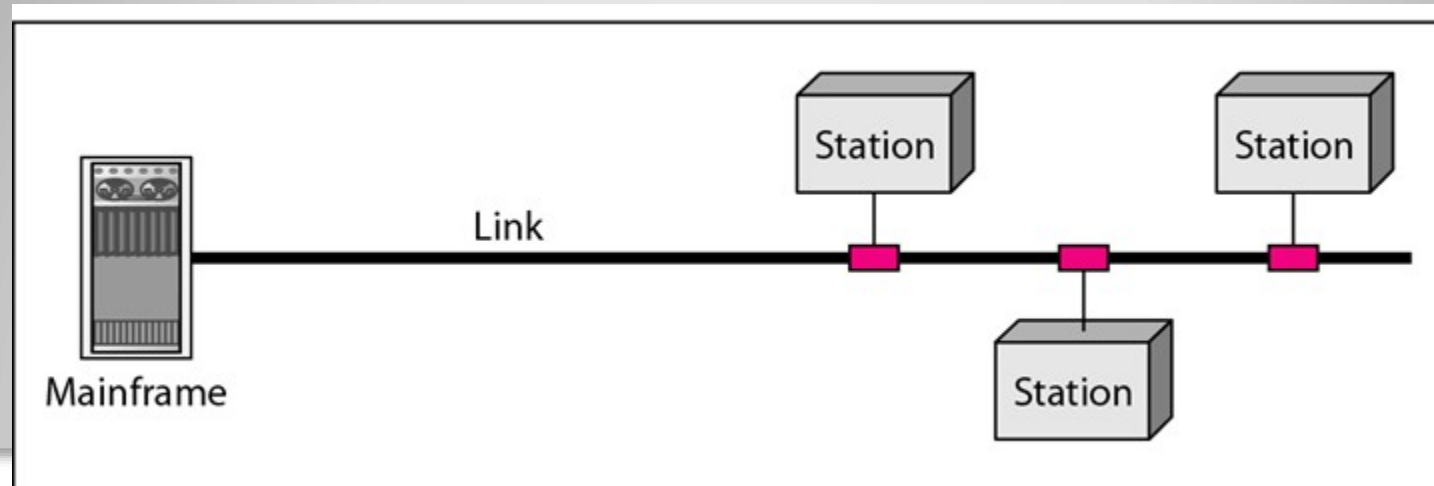
1. **Point to Point:** **A point-to-point connection provides a dedicated link between two devices.** The entire capacity of the link is reserved for transmission between those two devices. Most point-to-point connections use an actual length of wire or cable to connect the two ends.



Cont'd

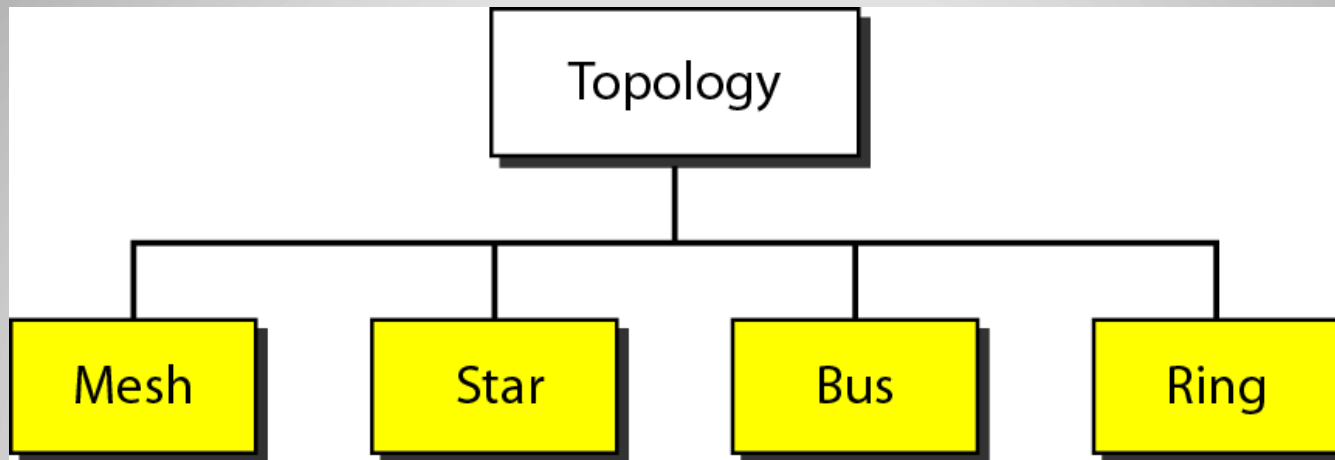
2. Multipoint: A multipoint (also called multidrop) connection is **one in which more than two specific devices share a single link.**

In a multipoint environment, the capacity of the channel is shared, **either spatially or temporally/timeshared.** **If several devices can use the link simultaneously, it is as spatially shared connection.** If users must take turns, it is a timeshared connection.



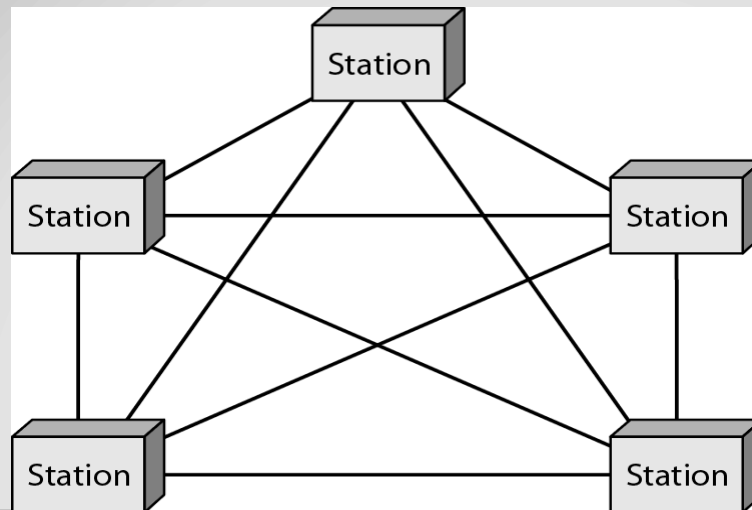
Topologies

- There are different types of network topologies;
 1. Mesh
 2. Star
 3. Bus
 4. Ring
 5. Hybrid



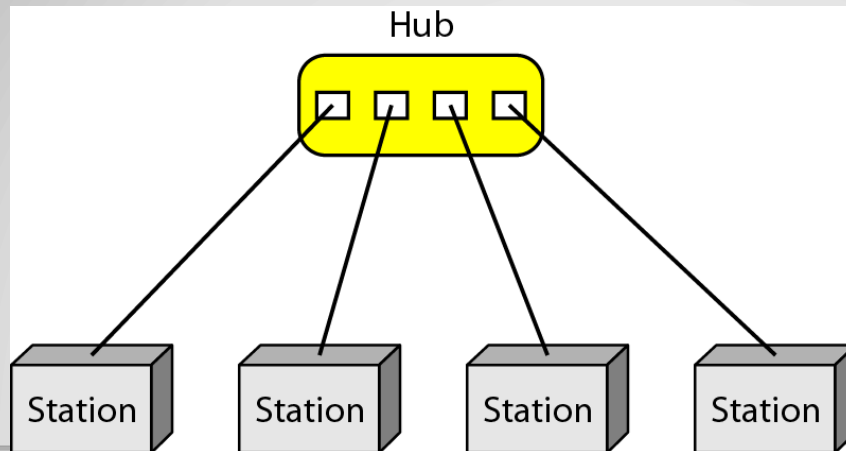
MESH

- In a mesh topology, **every device has a dedicated point-to-point link to every other device.**
- The term dedicated **means that the link carries traffic only between the two devices it connects**



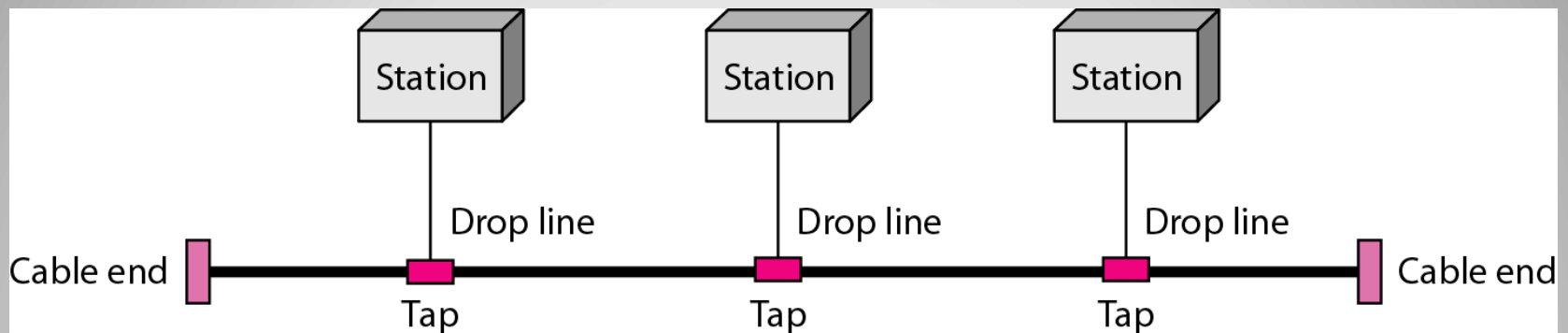
Star

- In a star topology, **each device has a dedicated point-to-point link only to a central controller, usually called a hub.**
- The devices are not directly linked to one another. Unlike a mesh topology, **a star topology does not allow direct traffic between devices.**
- The controller acts as an exchange: **If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.**



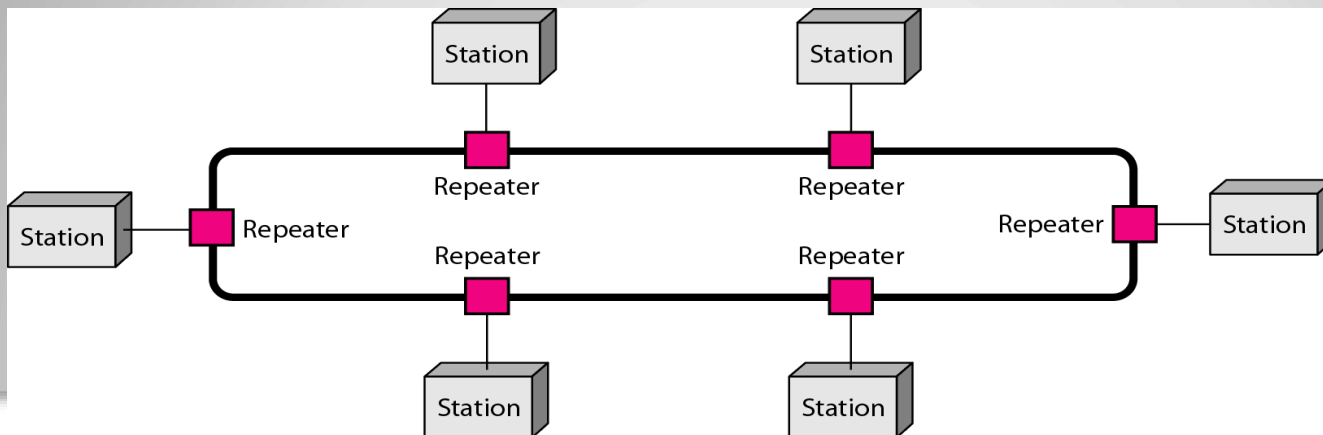
Bus

- The previous examples are all **point-to-point connections**, **a bus topology, on the other hand, is multipoint.**
- **One long cable acts as a backbone to link all the devices in a network, nodes are connected to the bus cable by drop lines and taps.**
- **A drop line is a connection running between the device and the main cable.**
- **A tap is a connector that either splices into the main cable or penetrates the sheathing of a cable to create a contact with the metallic core**



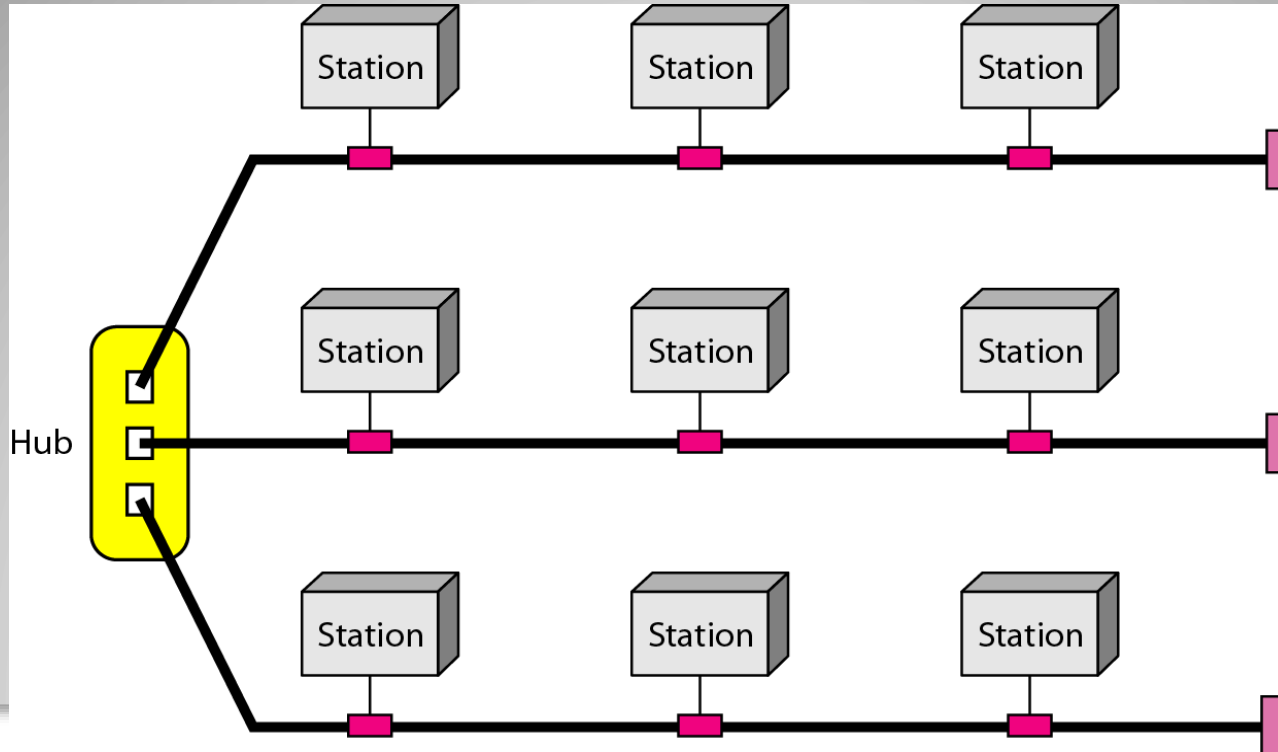
RING

- In a ring topology, **each device has a dedicated point-to-point connection with only the two devices on either side of it.**
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater. When a device **receives a signal intended for another device, its repeater regenerates the bits and passes them along.**



Hybrid (a star backbone with three bus networks)

- A network can be hybrid. For example, **we can have a main star topology with each branch connecting several stations in a bus topology**



Protocols and Standards

Protocols

- **A protocol is identical with a rule.**
- It consists of a set of rules that govern data communications.
- **It determines what is communicated, how it is communicated and when it is communicated.**
- The key elements of a protocol are **syntax, semantics and timing**

Elements of a Protocol

1. Syntax: The term syntax refers to the structure or format of the data, meaning the order in which they are presented. For example, a simple protocol might expect the first 8bits of data to be the address of the sender, the second 8bits to be the address Of the receiver, and the rest of the stream to be the message itself.

2. Semantics: The word semantics refers to the meaning of each section of bits. How is a particular pattern to be interpreted, and what action is to be taken based on that interpretation? For example, does an address identify the route to be taken or the final destination of the message?

3. Timing: The term timing refers to two characteristics: when data should be sent and how fast they can be sent. For example, if a sender produces data at 100 Mbps But the receiver can process data at only 1Mbps, the transmission will overload the receiver and some data will be lost.

Standards

- **Standards are agreed-upon rules.**
- Standards are essential in **creating and maintaining an open and competitive market for equipment manufacturers** and **in guaranteeing national and international interoperability of data and telecommunications technology and processes.**
- Data communication standards fall into two categories: **de facto (meaning "by fact" or "by convention")** and **de jure (meaning "by law" or "by regulation")**.

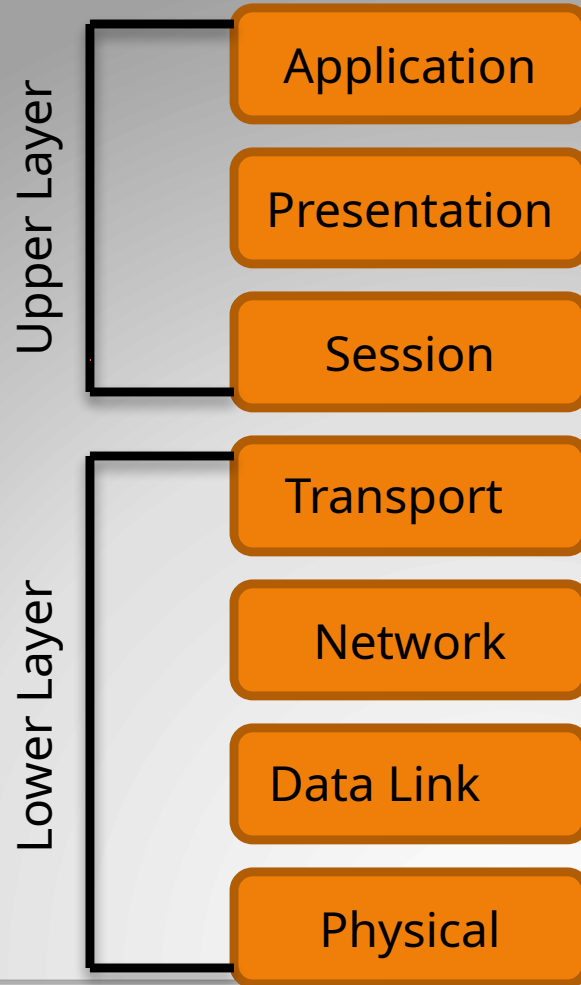
Standards Organizations

- Standards are developed through the **cooperation of standards creation committees, forums, and government regulatory agencies.**
- There are different standard creation committees;
 1. International Organization for Standardization(ISO)
 2. International Telecommunication Union-Telecommunication Standards Sector (ITU-T).
 3. American National Standards Institute (ANSI).
 4. Institute of Electrical and Electronics Engineers (IEEE).
 5. Electronic Industries Association (EIA).

ISO / OSI model

- Established in 1947, the International Standards Organization (**ISO**) is **a multinational body dedicated to worldwide agreement on international standards.**
- An ISO standard that **covers all aspects of network communications is the Open Systems Interconnection (OSI) model.** It was first introduced in the late 1970s.
- Note: **ISO is the organization. OSI is the model**

Layers of the OSI Model

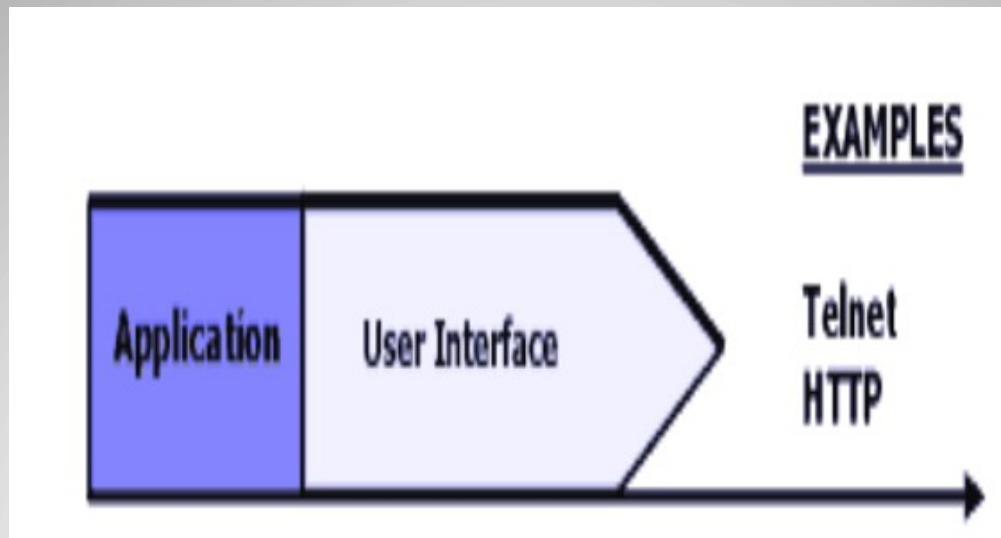


Upper layers provides the way applications with in the end stations will communicate with each other and with users

Lower Layers provides the ability for end stations to establish connections to each other in order to exchange information

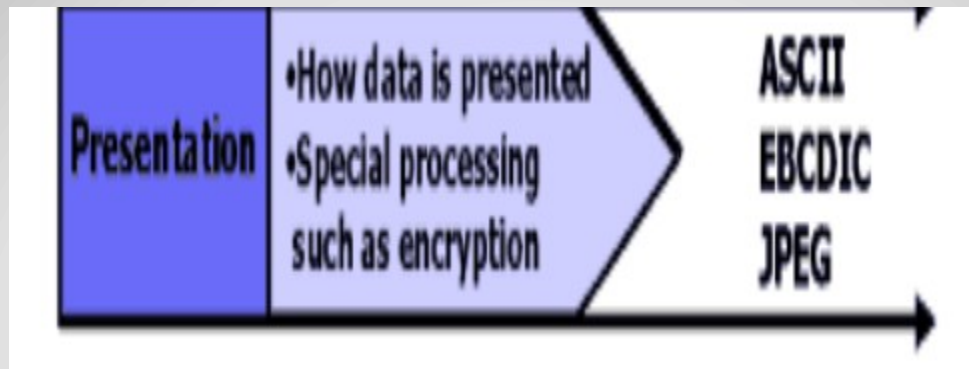
Application Layer

- **The application layer contains a variety of protocols that are commonly needed by users. One widely-used application protocol is HTTP (HyperText Transfer Protocol),** which is the basis for the World Wide Web.



Presentation Layer

- The presentation layer is concerned with the syntax and semantics of the information transmitted..



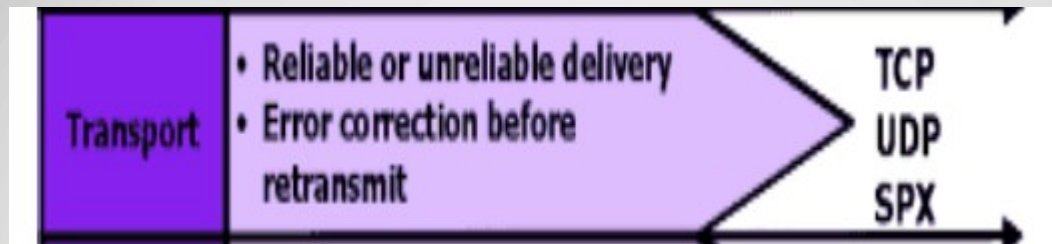
Session Layer

- **The session layer allows users on different machines to establish sessions between them.**
- Sessions offer various services, **including dialog control (keeping track of whose turn it is to transmit), token management (preventing two parties from attempting the same critical operation at the same time), and synchronization** (check pointing long transmissions to allow them to continue from where they were after a crash).



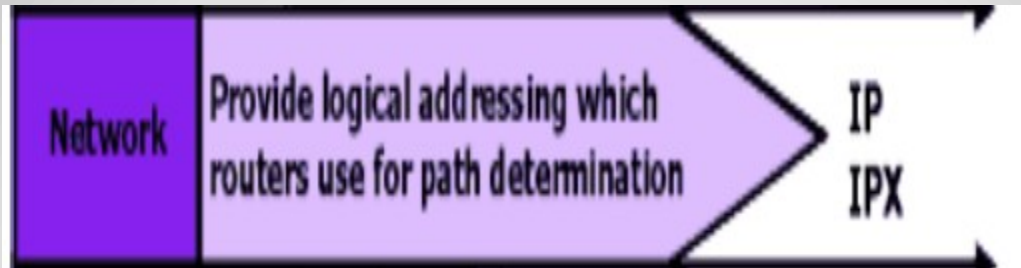
Transport Layer

- The basic function of the transport layer is to accept data from above, split it up into smaller units if need be, pass these to the network layer, and ensure that the pieces all arrive correctly at the other end.



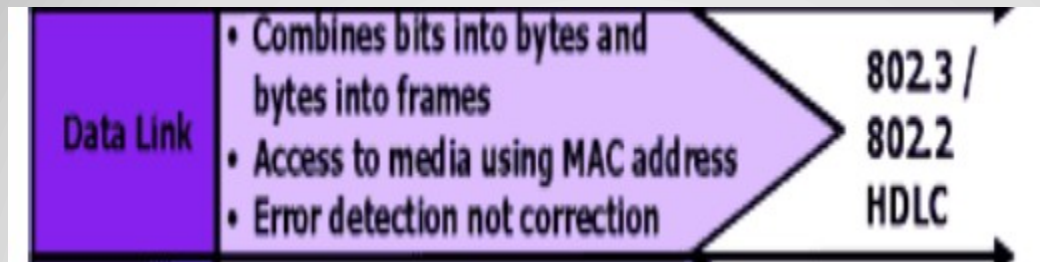
Network Layer

- **The network layer controls the operation of the subnet.**
-
- A key design issue is determining how packets are routed from source to destination.
- **Routes can be based on static tables that are "wired into" the network and rarely changed.**



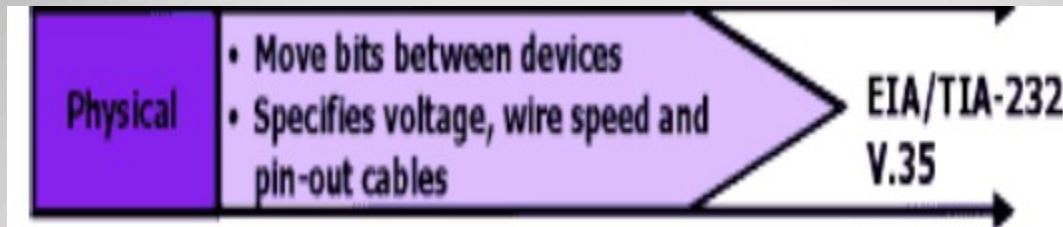
Data Link Layer

- The main task of the data link layer is to transform a raw transmission facility into a line that appears free of undetected transmission errors to the network layer.
- It accomplishes this task by having the sender break up the input data into data frames (typically a few hundred or a few thousand bytes) and transmit the frames sequentially.
- If the service is reliable, the receiver confirms correct receipt of each frame by sending back an acknowledgement frame



Physical Layer

- The physical layer is concerned with transmitting raw bits over a communication channel.
- The design issues have to do with making sure that when one side sends a 1 bit, it is received by the other side as a 1 bit, not as a 0 bit.



Practical Lab

- Labs will be simulated using Packet Tracer
- Getting Familiar with the simulator
- Static Routing Configurations (Lab 1)